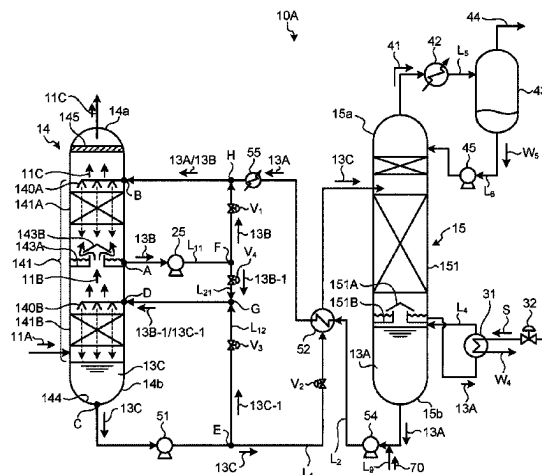


(45) **Date of Patent:** **Sep. 2, 2025**

(Continued)



section; and an absorbent discharge line that discharges a part of the first circulation solution from the first absorbent circulation line and supply the part of the first circulation solution as a discharged solution to the second absorbent circulation section.

6 Claims, 9 Drawing Sheets

(51) **Int. Cl.**

B01D 53/62 (2006.01)

B01D 53/78 (2006.01)

(52) **U.S. Cl.**

CPC **B01D 53/18** (2013.01); **B01D 53/62** (2013.01); **B01D 53/78** (2013.01); **B01D 2252/20478** (2013.01); **B01D 2257/504** (2013.01)

(58) **Field of Classification Search**

CPC B01D 53/78; B01D 2252/20478; B01D

2257/504; B01D 53/1475; Y02C 20/40;

Y02P 20/10; Y02P 20/151; C01B 32/50

See application file for complete search history.

(56)

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FIG. 1

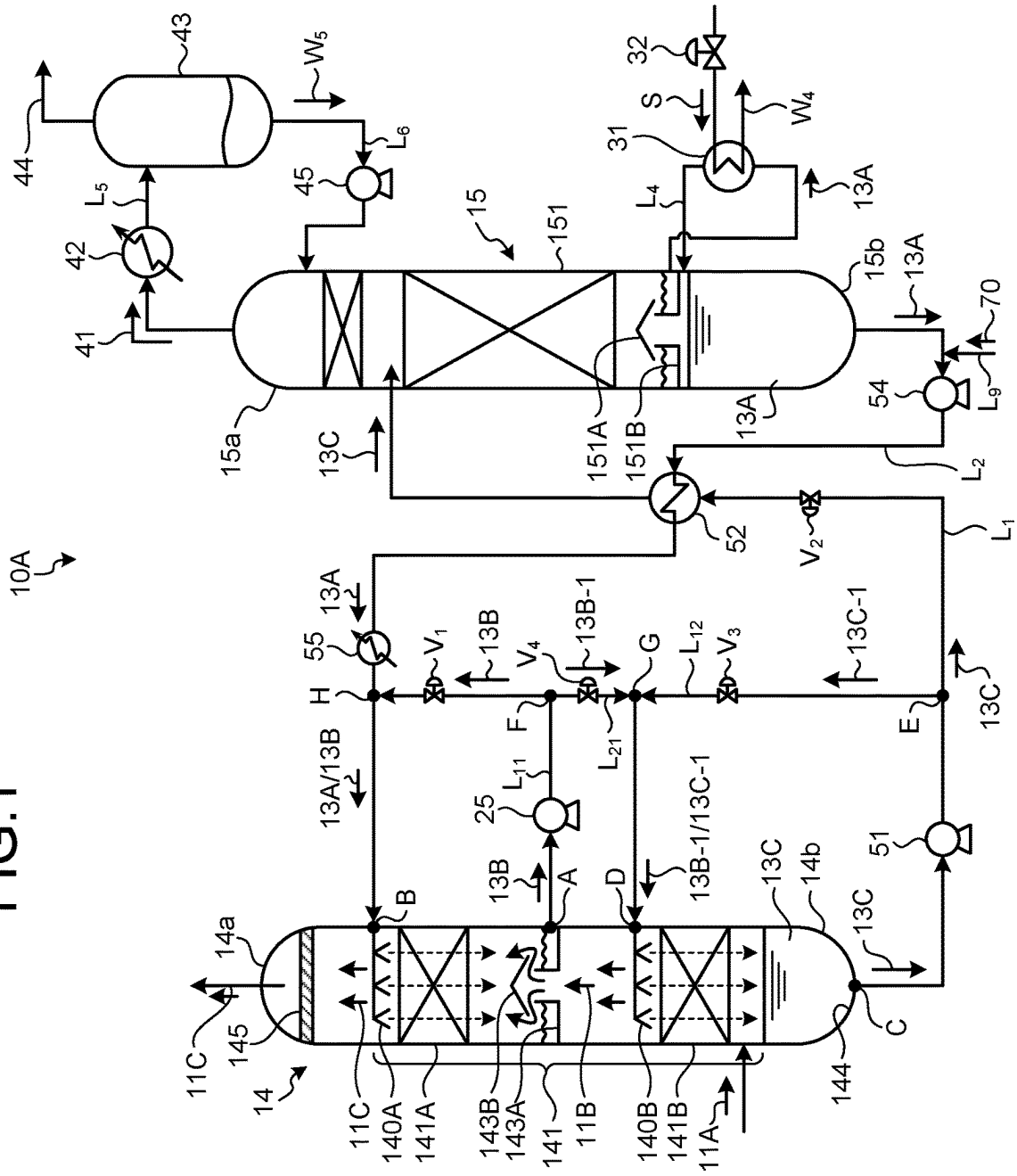


FIG. 2

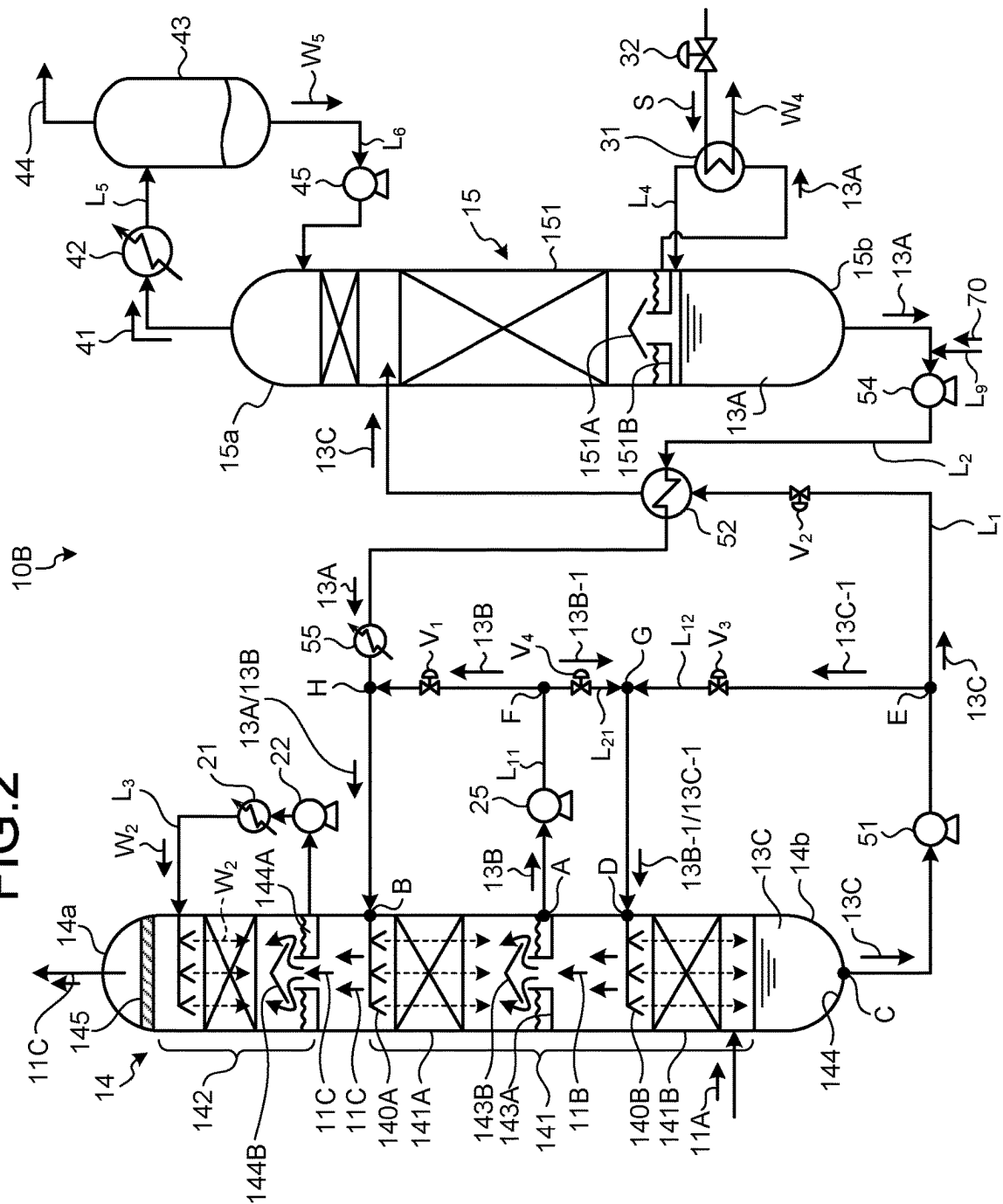


FIG. 3

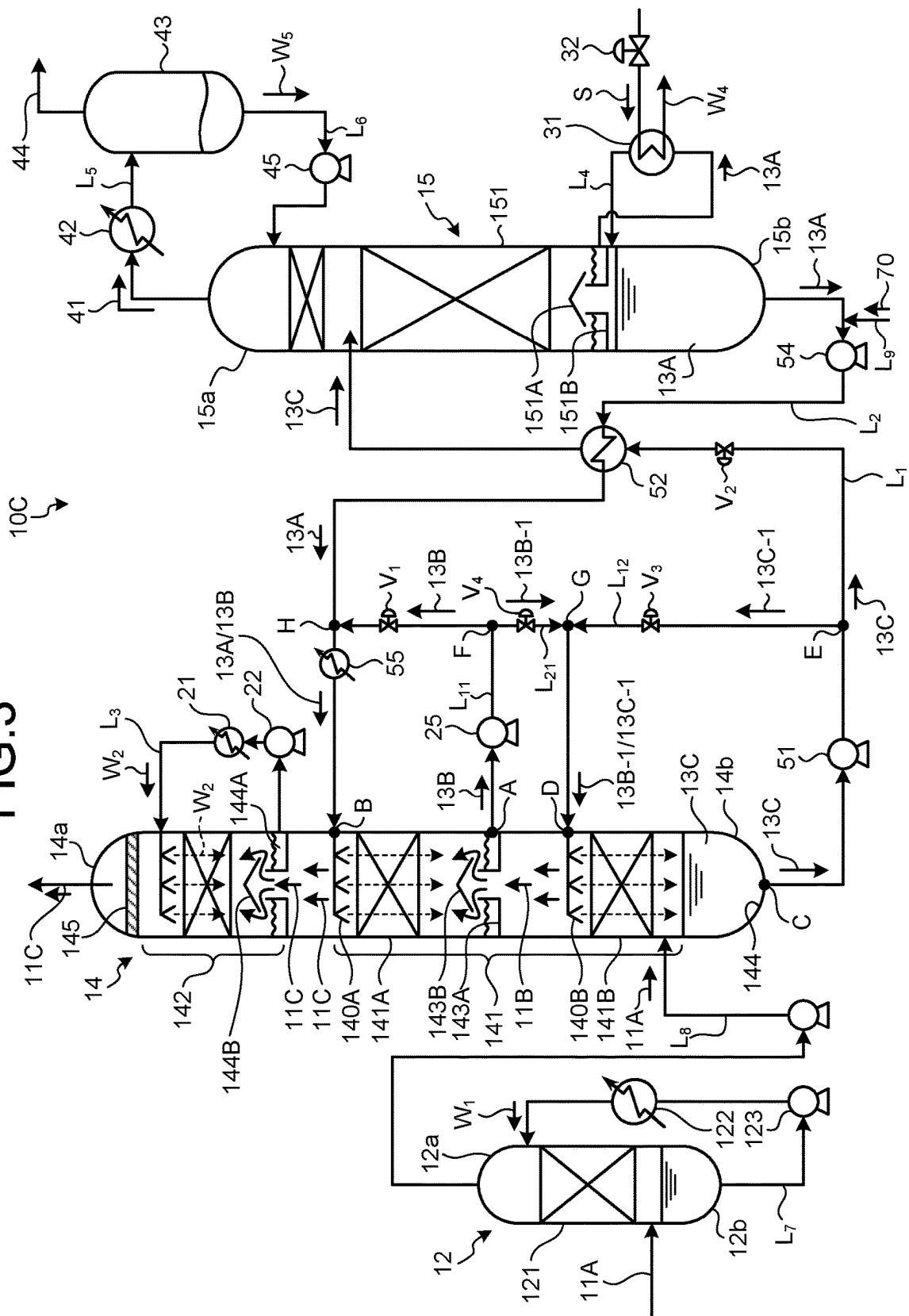


FIG.4

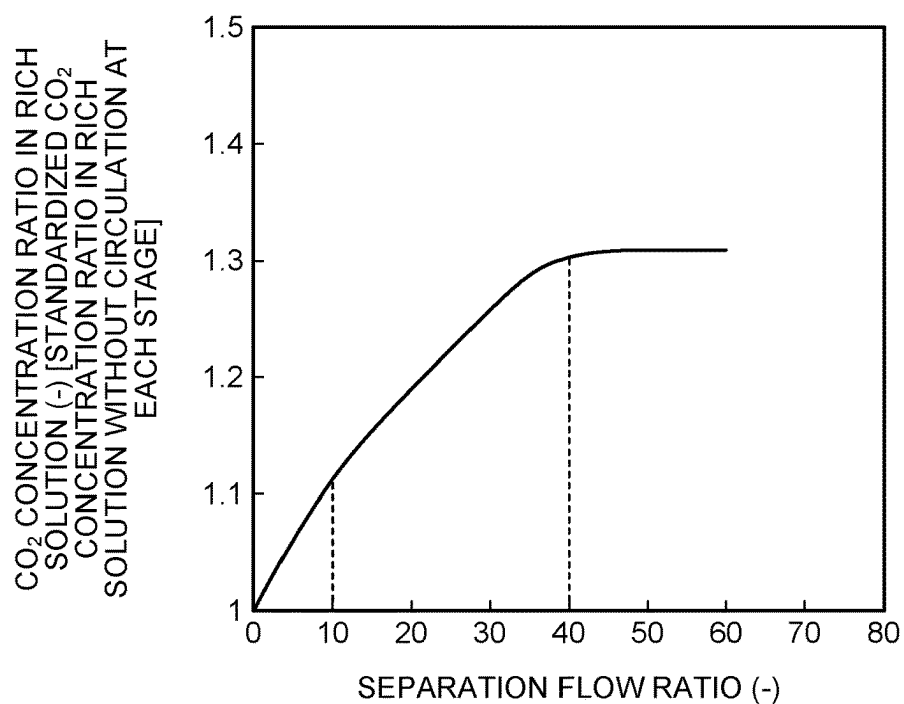


FIG.5

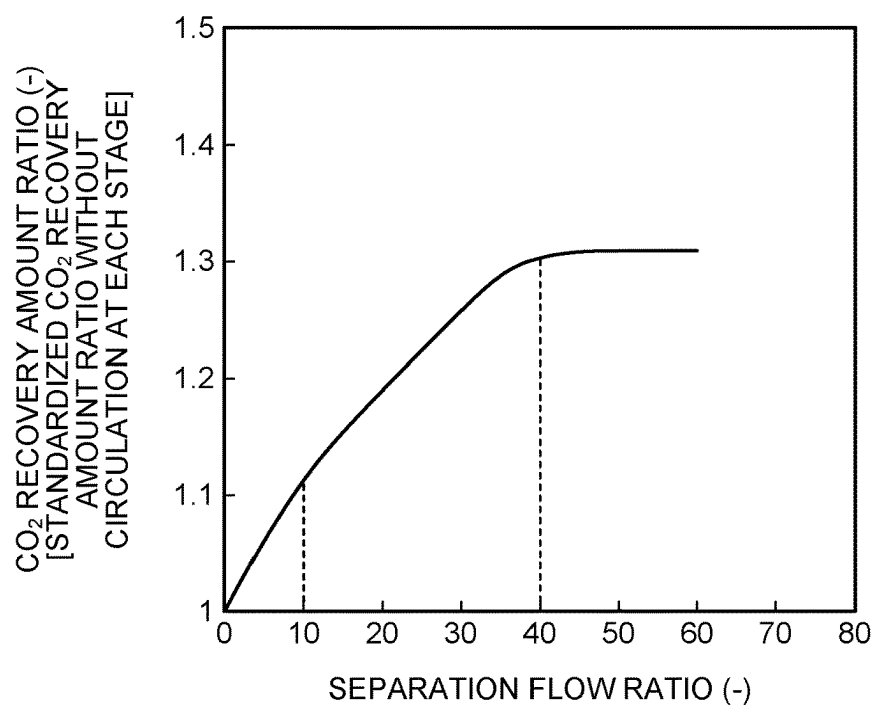


FIG.6

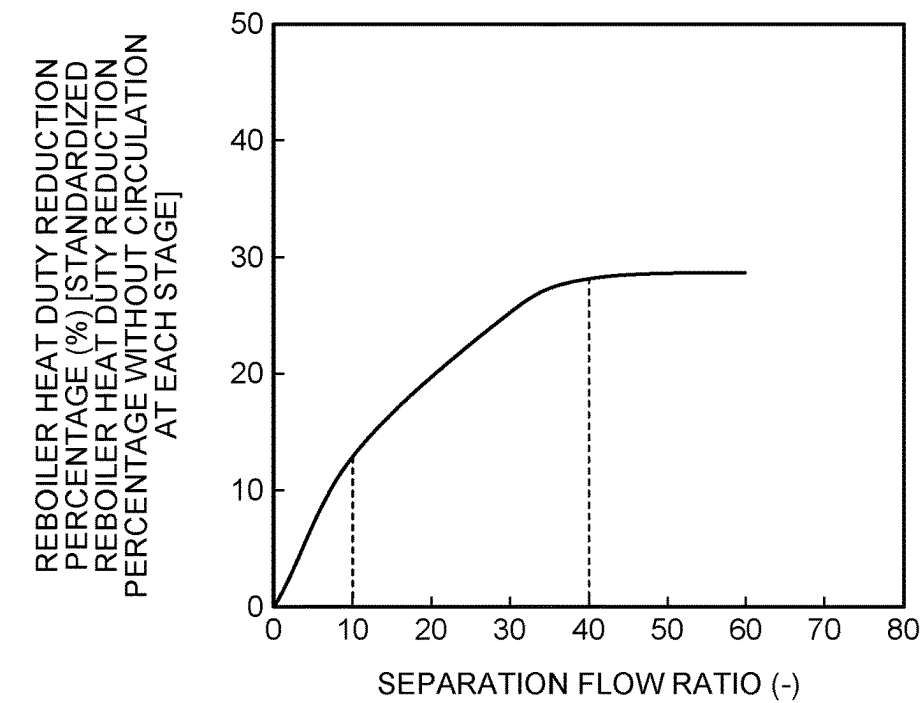


FIG.7

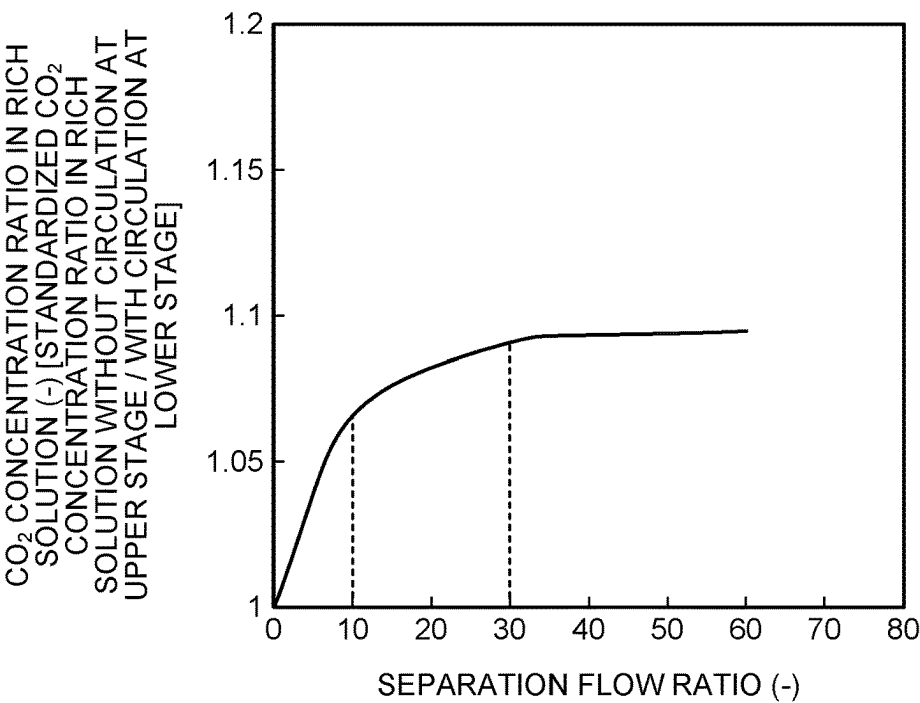


FIG.8

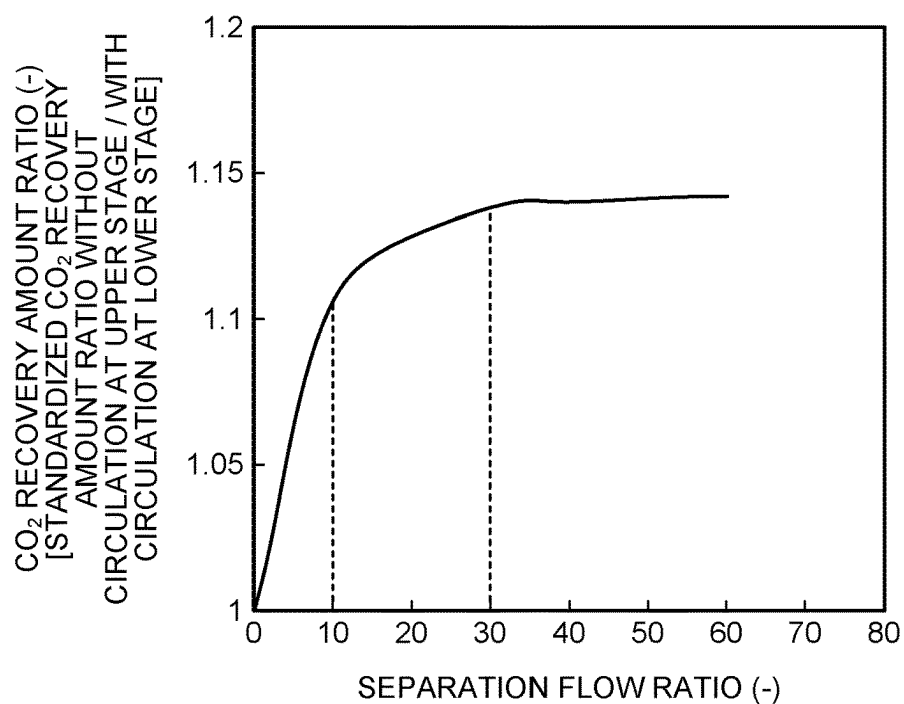


FIG.9

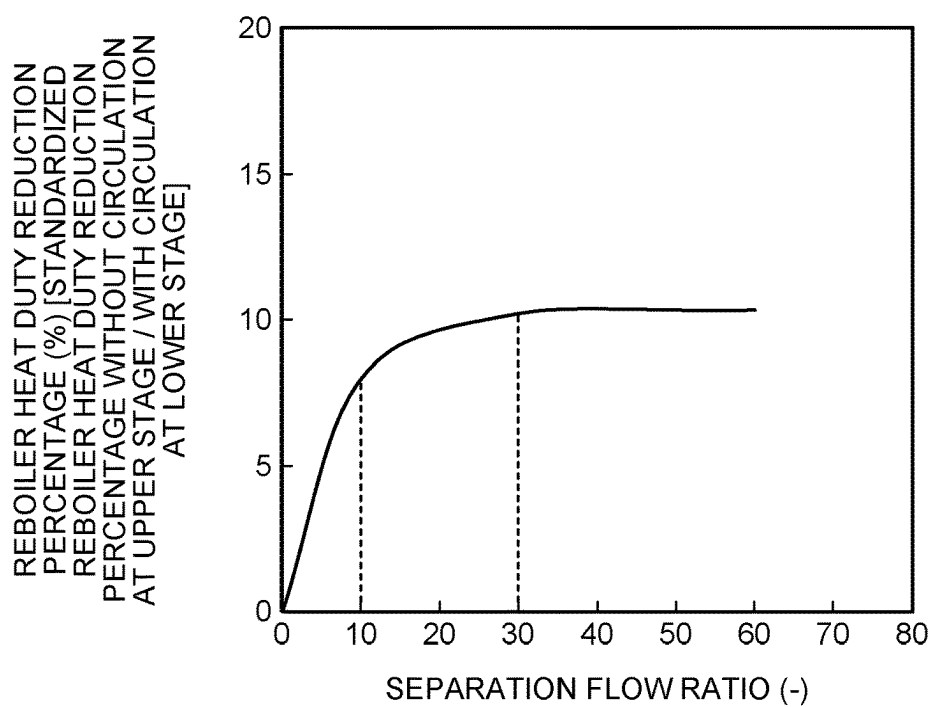


FIG.10

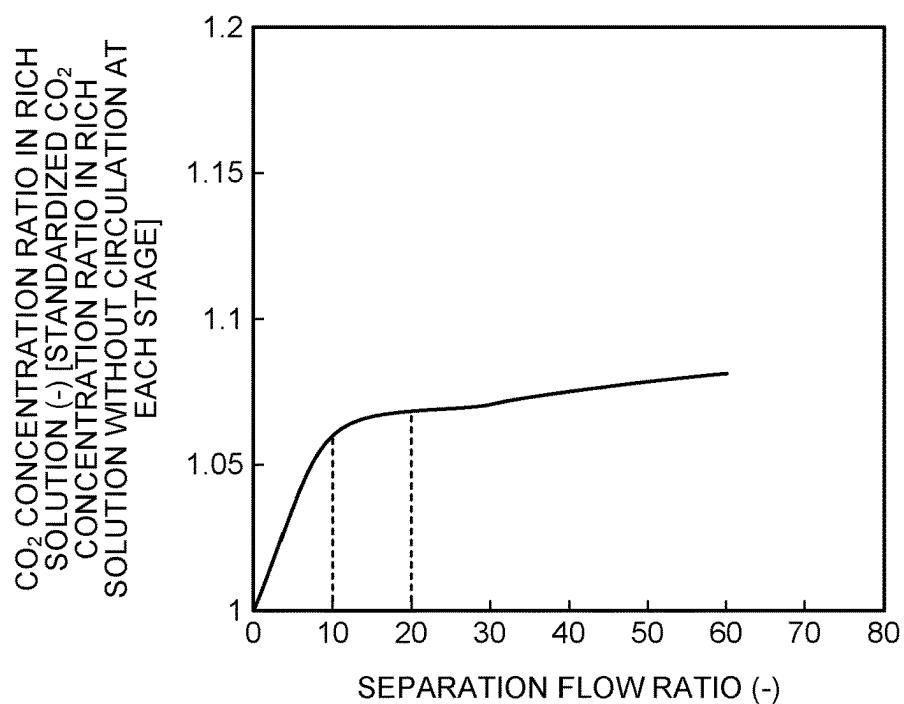


FIG.11

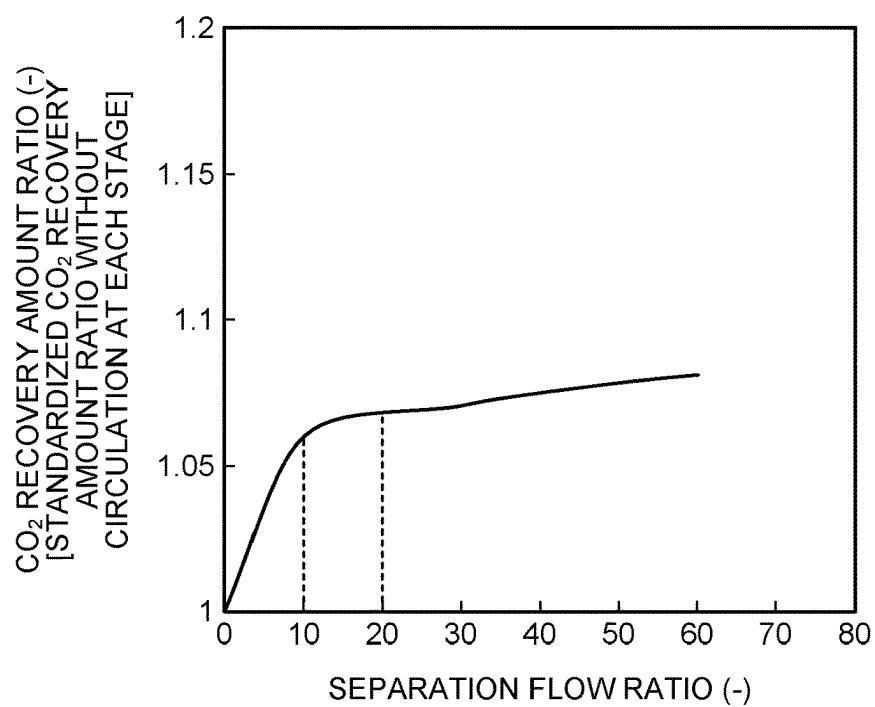


FIG.12

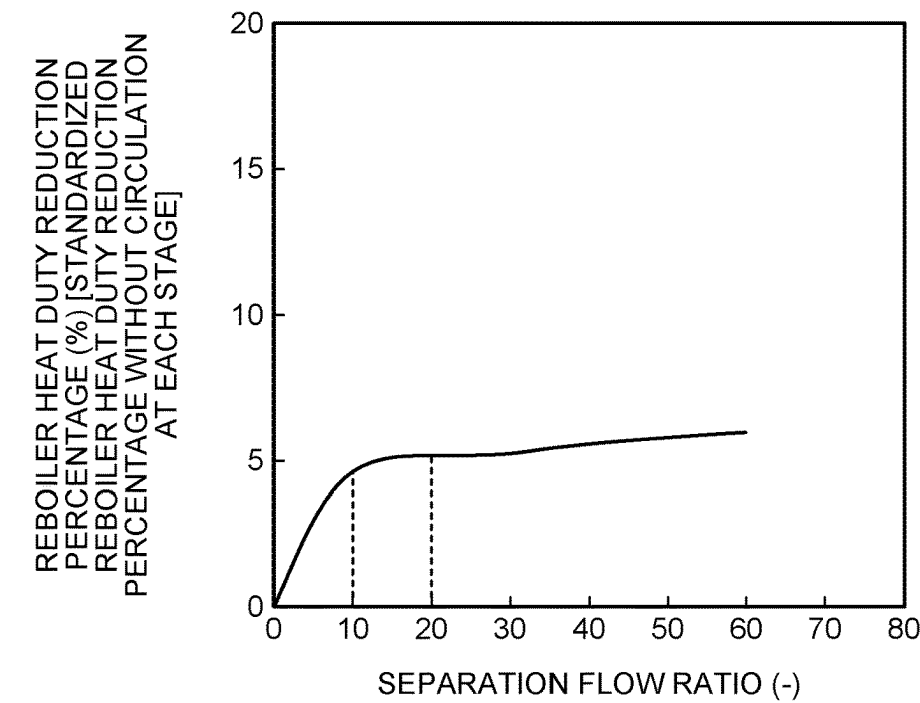


FIG.13

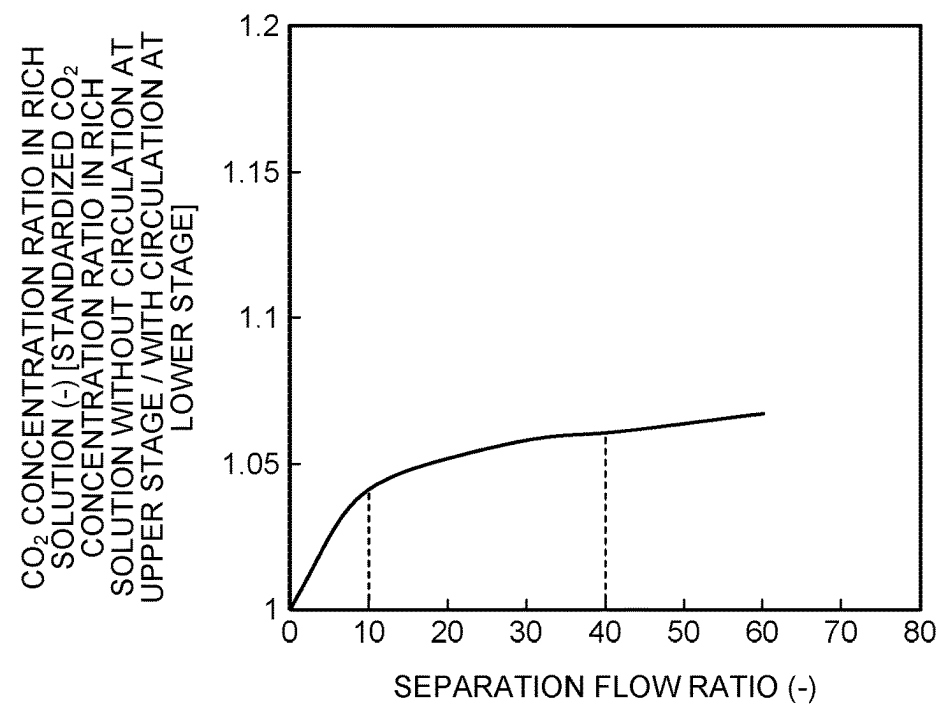


FIG.14

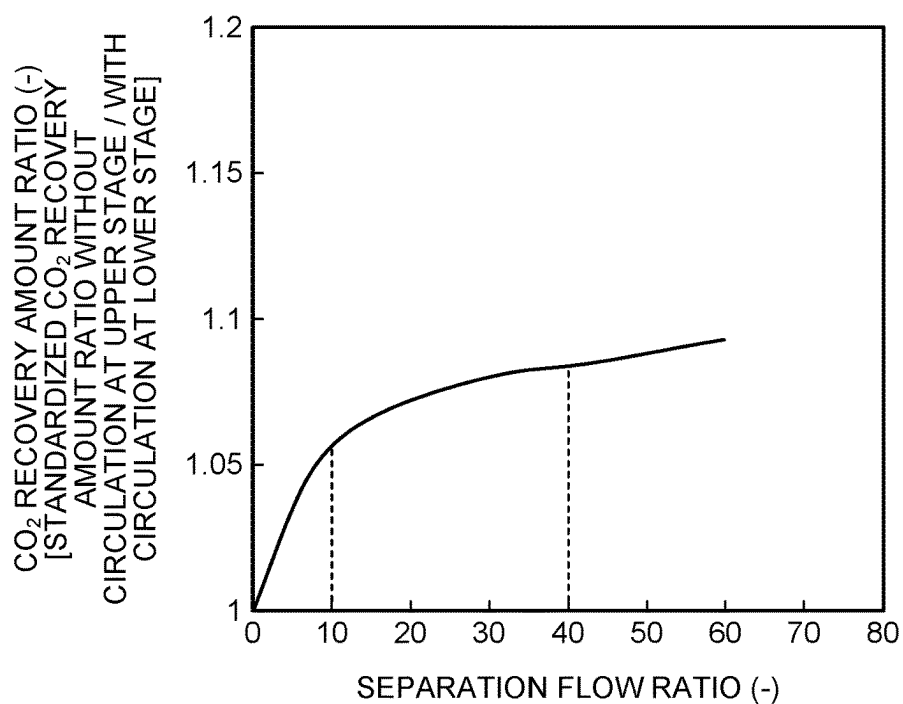
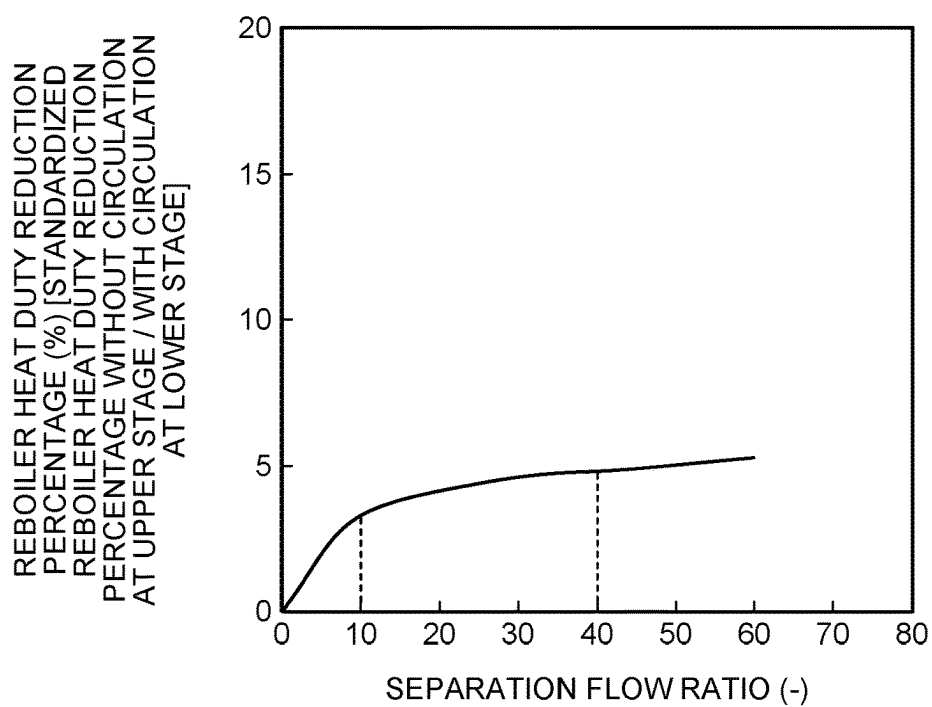


FIG.15



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CO₂ RECOVERY UNIT AND CO₂ RECOVERY METHOD**FIELD**

The present invention relates to a CO₂ recovery unit and a CO₂ recovery method.

BACKGROUND

Various methods have been proposed so far to recover and remove acid gases, especially CO₂, contained in flue gases from boilers, for example. One of such methods is to bring CO₂ into contact with a CO₂ absorbent made of an amine aqueous solution to remove and recover CO₂ contained in the flue gases.

For example, the method involves use of the CO₂ absorbent in a CO₂ absorber to absorb CO₂ contained in the flue gases for removal of CO₂ from the flue gases, followed by regeneration of the CO₂ absorbent by releasing CO₂ absorbed in the CO₂ absorbent and regenerate the CO₂ absorbent in a regenerator, and circulation of the regenerated CO₂ absorbent into the CO₂ absorber for reuse of the CO₂ absorbent to remove CO₂ from the flue gases again. The CO₂-absorbing CO₂ absorbent is heated in the regenerator with use of steam flowed from a reboiler to release CO₂ for recovery of CO₂ at a high purity. In this method, it is proposed that a part of the CO₂ absorbent is discharged at a lower part and returned to at least one portion(s) between an upper CO₂ absorbent supply part and a lower gas supply part in the CO₂ absorber to increase an amount of CO₂ absorbed from the flue gas while reducing a reboiler heat duty required for the regeneration of the CO₂ absorbent (for example, Patent Literature 1). It is proposed that the CO₂ absorber is provided at its absorption section with a plurality of stages to allow a part of a discharged solution to be returned to an upper side above a position for discharging a circulated solution at the lower stage in the absorption section to increase the amount of CO₂ absorbed from the flue gas while reducing an energy required for the regeneration of the CO₂ absorbent, as well (for example, Patent Literature 2).

CITATION LIST**Patent Literature**

Patent Literature 1: Japanese Translation of PCT Application Publication No. 2013-512088

Patent Literature 2: International Publication No. 09/104744

SUMMARY**Technical Problem**

However, the current CO₂ recovery unit has a problem of requiring a large amount of heat in reboiling for reuse of a CO₂ absorbent because of difficulties in CO₂ release at a CO₂ regenerator and reduction of steam energy required for regeneration of the CO₂ absorbent.

Furthermore, although the CO₂ content in a flue gas is 10 to 15% by volume (represented simply by “%”, hereinafter), there is a demand for a CO₂ recovery unit and a CO₂ recovery method that are capable of reducing the amount of heat in reboiling for the reuse of the CO₂ absorbent even when CO₂ is recovered from gases under nearly atmospheric

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pressure containing a subtle amount of CO₂ other than the flue gas as well as a variety of gases with low CO₂ contents in gases having CO₂ concentrations of 10% or less flowed from gas turbine gas and the like, for example.

In light of the aforementioned circumstances, the present invention has an object to provide a CO₂ recovery unit and a CO₂ recovery method that are capable of reducing an amount of heat in reboiling for reuse of a CO₂ absorbent to recover CO₂ from a variety of gases with low CO₂ contents.

Solution to Problem

A CO₂ recovery unit according to a first aspect of the present invention includes: a CO₂ absorber that brings a gas having a low CO₂ concentration into countercurrent contact with a CO₂ absorbent to remove CO₂; an absorbent regenerator that performs a heat exchange between a rich solution containing absorbed CO₂ and water vapor flowed from a reboiler to regenerate the CO₂ absorbent; a rich solution supply line that discharges a rich solution containing absorbed CO₂ from a bottom part of the CO₂ absorber and supplies the rich solution to an upper side of the absorbent regenerator; and a lean solution supply line that discharges a lean solution obtained by releasing CO₂ from a bottom part of the absorbent regenerator and supplies the lean solution to an upper side of the CO₂ absorber. The CO₂ absorber includes a CO₂ absorption section with at least two or more stages. The CO₂ recovery unit further includes: absorbent circulation lines that each discharge the CO₂ absorbent from a lower side of one of the stages in the CO₂ absorption section and supply the CO₂ absorbent as a circulation solution to an upper side of the corresponding stage used for the discharge in the CO₂ absorption section; and an absorbent discharge line that discharges a part of the circulation solution from the absorbent circulation line and supply the part of the circulation solution as a discharged solution to a stage immediately below the stage used for the discharge.

A CO₂ recovery method according to a second aspect of the present invention includes: a CO₂ absorption step of bringing a gas having a low CO₂ concentration into countercurrent contact with a CO₂ absorbent to remove CO₂; an absorbent regeneration step of performing a heat exchange between water vapor and a rich solution containing the absorbed CO₂ to regenerate the CO₂ absorbent; and a step of discharging a rich solution containing CO₂ that is absorbed at the CO₂ absorption step, supplying the rich solution to the absorbent regeneration step, discharging a lean solution that is obtained by releasing CO₂ from a tower bottom part at the absorbent regeneration step, and supplying the lean solution to the CO₂ absorption step to recover CO₂ in the gas. The CO₂ absorption step includes an absorbent circulation step of, in a CO₂ absorption section with at least two or more stages, discharging the CO₂ absorbent from lower sides of the stages in the CO₂ absorption section and supplying the CO₂ absorbent as a circulation solution to upper sides of the corresponding stages used for the discharge in the CO₂ absorption section; and an absorbent discharge step of discharging a part of the circulation solution from the absorbent circulation step and supplying the part of the circulation solution as a discharged solution to a stage immediately below the stage used for the discharge.

Advantageous Effects of Invention

According to the present invention, an amount of heat can be reduced in reboiling for CO₂ recovery to recover CO₂ from a variety of gases with low CO₂ contents.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a schematic view representing a configuration of a CO₂ recovery unit according to a first embodiment.

FIG. 2 is a schematic view representing a configuration of a CO₂ recovery unit according to a second embodiment.

FIG. 3 is a schematic view representing a configuration of a CO₂ recovery unit according to a third embodiment.

FIG. 4 is a graph representing CO₂ concentration ratios in a rich solution in test example 1.

FIG. 5 is a graph representing CO₂ recovery amount ratios in test example 1.

FIG. 6 is a graph representing reduction percentages (%) of reboiler heat duty in an absorbent regenerator in test example 1.

FIG. 7 is a graph representing CO₂ concentration ratios in a rich solution in test example 2.

FIG. 8 is a graph representing CO₂ recovery amount ratios in test example 2.

FIG. 9 is a graph representing reduction percentages (%) of reboiler heat duty in an absorbent regenerator in test example 2.

FIG. 10 is a graph representing CO₂ concentration ratios in a rich solution in test example 3.

FIG. 11 is a graph representing CO₂ recovery amount ratios in test example 3.

FIG. 12 is a graph representing reduction percentages (%) of reboiler heat duty in an absorbent regenerator in test example 3.

FIG. 13 is a graph representing CO₂ concentration ratios in a rich solution in test example 4.

FIG. 14 is a graph representing CO₂ recovery amount ratios in test example 4.

FIG. 15 is a graph representing reduction percentages (%) of reboiler heat duty in an absorbent regenerator in test example 4.

DESCRIPTION OF EMBODIMENTS

Hereinafter, preferred embodiments of the present invention are described in detail with reference to the accompanying drawings. The present invention is not limited to these embodiments, and modifications, additions, and omissions can be made by a person skilled in the art without departing from the spirit and scope stated in the claims.

First Embodiment

FIG. 1 is a schematic view representing a configuration of a CO₂ recovery unit according to a first embodiment. As illustrated in FIG. 1, a CO₂ recovery unit 10A is a unit for recovering CO₂ in an introduced gas (e.g., air) 11A under nearly atmospheric pressure with a low concentration of CO₂, for example. The CO₂ recovery unit 10A is provided with: a CO₂ absorber 14 configured to bring the introduced gas 11A into countercurrent contact with a lean solution 13A of a CO₂ absorbent for allowing CO₂ contained in the introduced gas 11A to be absorbed in the CO₂ absorbent and then removed; and an absorbent regenerator 15 that is formed at a subsequent stage of the CO₂ absorber 14 to release CO₂ from a rich solution 13C of CO₂ absorbent containing absorbed CO₂ and then regenerate the lean solution 13A.

In this CO₂ recovery unit 10A, the CO₂ absorbent is circulated between the CO₂ absorber 14 and the absorbent regenerator 15. The lean solution 13A of the CO₂ absorbent obtained as a result of releasing CO₂ absorbs CO₂ from the

gas in the CO₂ absorber 14 to become the rich solution 13C. This rich solution 13C is supplied to the absorbent regenerator 15. The supplied rich solution 13C release CO₂ in the absorbent regenerator 15, and is regenerated as the lean solution 13A to be subsequently supplied to the CO₂ absorber 14. Here, the CO₂ absorbent is a generic term referring to: the lean solution 13A obtained by releasing CO₂; a semi-rich solution 13B that absorbs a part of CO₂ in the gas; and the rich solution 13C that contains CO₂ absorbed from the gas and is discharged from the CO₂ absorber 14. Each of the names is selected depending on circulating positions in the CO₂ recovery unit 10A and the CO₂ content ratio.

The CO₂ absorbent usable in the present invention is not particularly limited. Examples of the CO₂ absorbent include amine compounds such as alkanolamines and hindered amines with alcoholic hydroxyl groups. Examples of such alkanolamines include monoethanolamine, diethanolamine, triethanolamine, methyl diethanolamine, diisopropanolamine, and diglycolamine. Among them, monoethanolamine (MEA) is generally preferred. Examples of such hindered amines with alcoholic hydroxyl groups include 2-amino-2-methyl-1-propanol (AMP), 2-(ethylamino)-ethanol (EAE), 2-(methylamino)-ethanol (MAE), and 2-(diethylamino)-ethanol (DEAE).

A rich solution supply line L₁ is provided to supply the rich solution 13C containing CO₂ absorbed in the CO₂ absorber 14 to an upper side of the absorbent regenerator 15 between a bottom 14b of the CO₂ absorber 14 and a top 15a of the absorbent regenerator 15. The rich solution supply line L₁ is provided with: a rich solution pump 51 configured to supply the rich solution 13C containing CO₂ absorbed in the CO₂ absorber 14 to the absorbent regenerator 15; and a rich/lean solution heat exchanger 52 configured to heat the rich solution 13C using the lean solution 13A that is obtained as a result of releasing CO₂ by heating in the absorbent regenerator 15.

A CO₂ absorption section 141 includes: a first CO₂ absorption stage 141A (referred to as “first absorption stage”, hereinafter) arranged to absorb CO₂ from the introduced gas 11A; and a second CO₂ absorption stage 141B (referred to as “second absorption stage”, hereinafter) located below the first absorption stage 141A that are disposed inside the CO₂ absorber 14 in its height direction.

A solution reservoir 143A and a chimney tray 143B are provided between the first absorption stage 141A and the second absorption stage 141B. The solution reservoir 143A is arranged to store the semi-rich solution 13B that is flowed down from upper portions of the first absorption stage 141A to reach lower portions of the first absorption stage 141A.

The solution reservoir 143A is provided with a first absorbent circulation line L₁₁ that is arranged to discharge a whole amount of the semi-rich solution 13B stored in the solution reservoir 143A through a discharge position A of the CO₂ absorber 14 and introduce the discharged solution into an introduction position B located at the upper portion of the first absorption stage 141A that is the same stage as that for the discharge.

The first absorbent circulation line L₁₁ is provided with a semi-rich solution pump 25 to circulate the semi-rich solution 13B into the upper portion of the first absorption stage 141A. The first absorbent circulation line L₁₁ is provided with a circulation flow control valve V₁ that controls a circulation flow rate of the circulated semi-rich solution 13B.

A front end of a lean solution supply line L₂ is connected to the first absorbent circulation line L₁₁ at a connection

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position H, such that the lean solution 13A regenerated in the absorbent regenerator 15 is mixed with the circulated semi-rich solution 13B (circulation solution 13B) to form a mixture (13A/13B). The mixture is introduced into an upper stage liquid distributor 140A. The front end of the lean solution supply line L_2 may be connected directly to the upper stage liquid distributor 140A at the upper portion of the first absorption stage 141A without connected to the first absorbent circulation line L_{11} such that the lean solution 13A and the circulated semi-rich solution 13B (circulation solution 13B) are individually introduced into the upper stage liquid distributor 140A.

A second absorbent circulation line L_{12} is provided to be connected at a liquid reservoir 144 located at a bottom of the CO_2 absorber to discharge a whole amount of the rich solution 13C stored in the liquid reservoir 144 through a discharge position C of the CO_2 absorber 14 and introduce the discharged solution into an introduction position D positioned at an upper part of the second absorption stage 141B that is the same stage as the stage for the discharge.

A part of the rich solution 13C is separated as a circulation solution 13C-1 at a discharge position E by the second absorbent circulation line L_{12} . The second absorbent circulation line L_{12} is connected at the discharge position E to a base end of the rich solution supply line L_1 . The rich solution 13C discharged is supplied through the rich solution supply line L_1 to the absorbent regenerator 15. The rich solution supply line L_1 is provided with a rich solution flow control valve V_2 to control the amount of the rich solution 13C supplied to the absorbent regenerator 15. The second absorbent circulation line L_{12} is provided with a circulation flow control valve V_3 to control the circulation flow rate of the circulated circulation solution 13C-1.

An absorbent discharge line L_{21} is provided to discharge a part of the semi-rich solution 13B (circulation solution 13B) circulated from the first absorbent circulation line L_{11} as a discharged solution 13B-1 at a discharge position F and supply the discharged solution 13B-1 to the second absorption stage 141B that is the adjacent stage immediately below the stage for the discharge.

In the present embodiment, the front end of the absorbent discharge line L_{21} is connected at a connection position G to the second absorbent circulation line L_{12} , such that the discharged solution 13B-1 is mixed with the circulation solution 13C-1 to form the mixture (13B-1/13C-1). The mixture is supplied through a single line to the introduction position D located at the upper portion of the second absorption stage 141B. The front end of the absorbent discharge line L_{21} may be connected directly to the lower stage liquid distributor 140B at the upper portion of the second absorption stage 141B without connected to the second absorbent circulation line L_{12} such that the discharged solution 13B-1 and the circulation solution 13C-1 are individually introduced into the lower stage liquid distributor 140B.

In addition, a discharge flow control valve V_4 is provided in the absorbent discharge line L_{21} to control the discharge flow rate of the discharged solution 13B-1.

In the first absorption stage 141A, the lean solution 13A regenerated in the absorbent regenerator 15 and the circulated semi-rich solution 13B (circulation liquid 13B) are supplied as the CO_2 absorbents. The supplied lean solution 13A and the circulated semi-rich solution 13B (circulation liquid 13B) are distributed from the upper stage liquid distributor 140A that is located at the upper portion of the first absorption stage 141A, and then flowed down within a packed bed. Here, the semi-rich solution 13B as the circu-

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lation solution has a higher CO_2 concentration than the lean solution 13A, but is still capable of absorbing CO_2 . Thus, the semi-rich solution 13B can be reused in the first absorption stage 141A to absorb CO_2 from the introduced gas 11B.

To the second absorption stage 141B, the circulation solution 13C-1 is supplied together with the discharged solution 13B-1 of the semi-rich solution 13B containing CO_2 absorbed partially from the introduced gas 11B in the first absorption stage 141A. The supplied discharged solution 13B-1 and the circulation solution 13C-1 are distributed from the lower stage liquid distributor 140B at the upper portions of the second absorption stage 141B and flowed down within the packed bed. Here, the rich solution 13C as the circulation solution has a higher CO_2 concentration than the semi-rich solution 13B, but is still capable of absorbing CO_2 . Thus, the rich solution 13C can be reused in the second absorption stage 141B to absorb CO_2 from the introduced gas 11A.

In the first absorption stage 141A of the CO_2 absorber 14, the introduced gas 11B containing CO_2 is brought into countercurrent contact with the mixture of the lean solution 13A made of the amine compound-based CO_2 absorbent and the circulated semi-rich solution 13B (circulation solution 13B) that are introduced into the tower. As a result, the gas introduced into the tower becomes an outlet gas 11C by removing CO_2 in the gas while the lean solution 13A becomes the semi-rich solution 13B.

In the second absorption stage 141B, the semi-rich solution 13B (discharged solution 13B-1 and the circulation solution 13C-1) partially absorbing CO_2 is brought into countercurrent contact with the introduced gas 11A that contains CO_2 and is introduced from a bottom side of the tower, allowing CO_2 to be absorbed into the semi-rich solution 13B from the introduced gas 11A by a chemical reaction. As a result, the introduced gas 11A becomes the introduced gas 11B with a reduced CO_2 concentration by removing CO_2 in the gas while the semi-rich solution 13B becomes the rich solution 13C with higher amount of absorbed CO_2 . This arrangement allows the introduced gases 11A and 11B containing CO_2 to be flowed through the first absorption stage 141A and the second absorption stage 141B, removing CO_2 to provide a decarbonated outlet gas 11C.

A mist eliminator 145 is installed inside a tower top 14a to capture mist contained in the decarbonated outlet gas 11C and then discharge the gas outwardly from the tower top 14a of the CO_2 absorber 14.

The absorbent regenerator 15 is provided at its interior with a packed bed 151 to release CO_2 with aid of water vapor from the rich solution 13C containing absorbed CO_2 . The absorbent regenerator 15 is provided at the vicinity of a tower bottom 15b with a circulation line L_4 that circulates therethrough a part of the lean solution 13A flowed down to the tower bottom 15b. The circulation line L_4 is provided with a reboiler 31 that generates water vapor by indirect heating of the lean solution 13A with use of saturated steam S, and a control valve 32 that controls the amount of saturated steam S supplied to the reboiler 31. The saturated steam S becomes steam condensate W_4 after the heating.

The absorbent regenerator 15 is provided at its top 15a with a gas discharge line L_5 to discharge CO_2 gas 41 together with water vapor. The gas discharge line L_5 is provided with a condenser 42 that condenses the water vapor discharged together with the CO_2 gas 41 into water, and a separation drum 43 to separate a CO_2 gas 44 from condensed water W_5 . The CO_2 gas 44 separated from the condensed water W_5 is discharged outwardly from an upper portion of the separa-

tion drum 43. A condensed water line L_6 is provided between the bottom part of the separation drum 43 and the top 15a of the absorbent regenerator 15 to supply the condensed water W_5 separated in the separation drum 43 to the top 15a of the absorbent regenerator 15. The condensed water line L_6 is provided with a condensed water circulation pump 45 to supply the condensed water W_5 separated in the separation drum 43 to the top 15a of the absorbent regenerator 15.

The lean solution supply line L_2 is provided between the bottom 15b of the absorbent regenerator 15 and the top 14a of the CO_2 absorber 14 to supply the lean solution 13A of the CO_2 absorber from the bottom 15b of the absorbent regenerator 15 to the upper side of the CO_2 absorption section 141. The lean solution supply line L_2 is provided with: a rich/lean solution heat exchanger 52 that heats the rich solution 13C containing the absorbed CO_2 with use of the lean solution 13A that is obtained by heating with water vapor in the absorbent regenerator 15 for the removal of CO_2 ; a lean solution pump 54 that supplies the lean solution 13A from the bottom 15b of the absorbent regenerator 15 to the upper portion of the CO_2 absorption section 141; and a cooling part 55 that cools the lean solution 13A of the CO_2 absorber down to a predetermined temperature.

In the present embodiment, this cooling part 55 is configured to cool the lean solution 13A down to the same temperature as or a temperature close to a gas temperature (room temperature: e.g., 25° C.) of the introduced gas 11A introduced into the CO_2 absorber 14.

Next, the overall operation of the CO_2 recovery unit 10A according to the present embodiment is described. For example, the introduced gas (atmospheric gas) 11A containing a subtle amount of CO_2 is introduced into the interior of the CO_2 absorber 14 through the bottom 14b of the CO_2 absorber 14 and flows upward inside the tower. The gas introduced into the CO_2 absorber 14 is brought into countercurrent contact with the CO_2 absorbent containing the amine compound such as alkanolamine in the first absorption stage 141A and second absorption stage 141B of the CO_2 absorption section 141, such that CO_2 is absorbed from the gas into the CO_2 absorbent and then removed for the purpose of providing the decarbonated outlet gas 11C.

The decarbonated outlet gas 11C is discharged outwardly from the top 14a of the CO_2 absorber 14.

After the absorption of CO_2 in the CO_2 absorber 14, the rich solution 13C of the CO_2 absorbent is flowed in the rich solution supply line L_1 to pass through the rich/lean solution heat exchanger 52 for heat exchange with the lean solution 13A and then supplied to the top side of the absorbent regenerator 15 with the aid of the rich solution pump 51.

The rich solution 13C of the CO_2 absorbent supplied to the absorbent regenerator 15 is allowed to release CO_2 by the heat of the water vapor supplied from the reboiler 31 while flowing downward inside the packed bed 151 of the absorbent regenerator 15, so as to become the lean solution 13A. In the absorbent regenerator 15, the solution is circulated from the liquid reservoirs 151B of the chimney tray 151A into the circulation line L_4 while heated by saturated steam S in the reboiler 31 to generate water vapor inside the absorbent regenerator 15. The lean solution 13A is obtained as a result of releasing CO_2 from the rich solution 13C with the aid of the heat of this generated water vapor. After the heating, the saturated steam S becomes steam condensate W_4 . The CO_2 gas 41 released together with the water vapor from the CO_2 absorbent is cooled down for the condensation into water with use of the condenser 42, separated from the condensed water W_5 , and then discharged as the CO_2 gas 44

outwardly through the upper portion of the separation drum 43. The condensed water W_5 separated is supplied to the absorbent regenerator 15.

After discharged from the bottom 15b of the absorbent regenerator 15, the lean solution 13A is flowed in the lean solution supply line L_2 to pass through the rich/lean solution heat exchanger 52 for heat exchange with the rich solution 13C and then supplied to the upper side of the CO_2 absorption section 141 of the CO_2 absorber 14 with the aid of the lean solution pump 54.

After supplied to the upper side of the CO_2 absorption section 141, the lean solution 13A absorbs CO_2 from the introduced gas 11B in the first absorption stage 141A to become the semi-rich solution 13B, which is discharged at the discharge position A at the lower portion of the first absorption stage 141A and flowed into the first absorbent circulation line L_{11} . The discharged semi-rich solution 13B is supplied with the aid of the semi-rich solution pump 25 together with the lean solution 13A to the introduction position B positioned at the upper side of the first absorption stage 141A.

In the second absorption stage 141B, CO_2 is absorbed from the introduced gas 11B to form the rich solution 13C. The rich solution 13C is discharged at the discharge position C at the lower part of the second absorption stage 141B and flowed into the second absorbent circulation line L_{12} . The discharged rich solution 13C is separated at the discharge position E, as the circulation solution 13C-1. Then, the circulation solution 13C-1 is supplied with the aid of the rich solution pump 51 to the introduction position D at the upper portion of the second absorption stage 141B.

A part of the semi-rich solution 13B is separated at the discharge position F of the first absorbent circulation line L_{11} by the absorbent discharge line L_{21} . The discharged solution 13B-1 is mixed with the circulation solution 13C-1 at the connection position G of the second absorbent circulation line L_{12} . This discharged solution 13B-1 and the circulation solution 13C-1 are mixed with each other (13B-1/13C-1) and then supplied through a single line to the introduction position D at the upper portion of the second absorption stage 141B.

After separated from the circulation solution 13C-1 at the discharge position E, the rich solution 13C is supplied to the absorbent regenerator 15 through the rich solution supply line L_1 .

Therefore, the configuration according to the present embodiment includes: the first absorbent circulation line L_{11} for supplying the semi-rich solution 13B discharged at the lower side of the first absorption stage 141A in the CO_2 absorber 14 to the upper side of the same stage 141A as the stage for the discharge; the second absorbent circulation line L_{12} for supplying the rich solution 13C discharged at the lower side of the second absorption stage 141B of the CO_2 absorber 14 to the upper side of the second absorption stage 141B that is the same stage as the stage for the discharge; the absorbent discharge line L_{21} for separating the circulated semi-rich solution 13B (circulation solution 13B) discharged at the lower side of the first absorption stage 141A and then mixing the separated solution with the lean solution 13C-1 that is discharged at the lower side of the second absorption stage 141B that is the adjacent stage immediately below the stage for the discharge; the lean solution supply line L_2 for supplying the lean solution from the absorbent regenerator 15 to the upper portion of the first absorption stage 141A that is the highest stage of the CO_2 absorber 14 and then mixing the supplied solution with the semi-rich solution 13B discharged at the lower side of the first

absorption stage **141A**; and the rich solution supply line L_1 for flowing the rich solution **13C** from the bottom part of the CO_2 absorber **14** to the absorbent regenerator **15**.

Thus, according to the first embodiment, after absorbing CO_2 , the semi-rich solution **13B** and the rich solution **13C** that contain absorbed CO_2 can be discharged and flowed outwardly through the lower sides of the first absorption stage **141A** and the second absorption stage **14B**, respectively, and supplied as the circulation solutions to the upper portions at the corresponding stages above the absorbent-discharged portions. This allows the circulated circulation solutions to be reused as the absorbents in the first absorption stage **141A** and the second absorption stage **14B** to recover CO_2 in the gas. As a result, with increase of the CO_2 concentration in the CO_2 absorbent, it is possible to improve the CO_2 recovery efficiency in the CO_2 absorber **14** and thereby increase the CO_2 recovery amount per unit volume of the CO_2 absorbent. Furthermore, the CO_2 concentration in the rich solution **13C** can be increased at the bottom **14b** of the CO_2 absorber **14**, thereby allowing the rich solution **13C** having an increased CO_2 concentration to be supplied to the absorbent regenerator **15**.

Here, when the two-stage configuration for the absorption section, the first absorption stage **141A** and the second absorption stage **141B**, as in the present embodiment, the solution discharged from the circulation solution **13C-1** is the rich solution **13C**, while the rich solution supply line L_1 functions as an absorbent discharge line to supply the discharged rich solution **13C** to the absorbent regenerator **15**. In the case of three-stage configuration for the absorption section, the first absorption stage **141A**, the second absorption stage **141B**, and a third absorption stage, a solution discharged from the circulation solution **13C-1**, which is supplied from the second absorption stage **141B**, is supplied as the discharged solution to the third absorption stage immediately below the discharged stage of the second absorption stage.

As a result, the semi-rich solution **13B** and the rich solution **13C** are reused twice in the CO_2 absorber **14**, enabling it to reduce the flow rate of the circulated CO_2 absorbent in the system circulating between the CO_2 absorber **14** and the absorbent regenerator **15**. This can reduce the flow rate of the rich solution **13C** supplied to the absorbent regenerator **15**, reducing the steam supply amount of the reboiler **31** used in the absorbent regenerator **15** for the regeneration of the CO_2 absorbent, and thereby reducing heat energy consumption to improve energy efficiency.

In the present embodiment, even in the case of recovering CO_2 from a gas containing a very subtle amount of CO_2 (e.g., air), the CO_2 concentration can be increased in the rich solution **13C** supplied to the absorbent regenerator **15** so as to reduce the steam amount required in the absorbent regenerator **15**, thereby reducing the thermal energy as well as achieving further energy saving in the process of regenerating the CO_2 absorbent.

Next, the separation ratios of the circulation solution to the discharged solution in the CO_2 recovery unit **10A** according to the present embodiment, is described.

In the present embodiment, regarding the flow rate (f1) of the semi-rich solution **13B** (circulation solution **13B**) circulated through the first absorbent circulation line L_{11} from the lower portion of the first absorption stage **141A** and the flow rate (f2) of the discharged solution **13B-1** discharged via the absorbent discharge line L_{21} , the separation ratio of the flow rates (referred to as "separation flow ratio", hereinafter) is preferably 1:1 to 60:1, preferably 10:1 to 60:1, more preferably 30:1 to 60:1.

Regarding the flow rate (f3) of the circulation solution **13C-1** flowed into the second absorbent circulation line L_{12} and the flow rate (f4) of the rich solution **13C** flowed into the rich solution supply line L_1 from the bottom of the CO_2 absorber **14**, the flow separation ratio of the flow rates is preferably 1:1 to 60:1, preferably 10:1 to 60:1, more preferably 30:1 to 60:1.

Hereinafter, preferred test examples exhibiting effects of the separation flow ratio according to the present embodiment are described. The present invention is not limited to these examples. FIGS. **4** to **9** in these test examples illustrate examples for gas having a CO_2 concentration of 1% or less.

FIG. **4** is a graph representing CO_2 concentration ratios in the rich solution in test example 1. Herein, FIG. **4** represents a performance improvement ratio calculated based on standardized CO_2 concentration (1) without the discharge at each stage. In FIG. **4**, the horizontal axis represents the separation flow ratio of the flow rate (f1) of the semi-rich solution **13B** (circulation solution **13B**) circulated through the first absorbent circulation line L_{11} to the flow rate (f2) of the discharged solution **13B-1** supplied to the absorbent discharge line L_{21} , in the range of 1:1 to 60:1 (the same applies to FIGS. **5** and **6**). The same applies to the separation flow ratio of the flow rate (f1) of the circulation solution **13C-1** circulated through the second absorbent circulation line L_{12} to the flow rate (f2) of the rich solution **13C** supplied to the rich solution supply line L_1 , in the range of 1:1 to 60:1.

FIG. **5** is a graph representing CO_2 recovery amount ratios in test example 1. Herein, FIG. **5** represents a performance improvement ratio calculated based on standardized CO_2 recovery amount (1) without the discharge at each stage. FIG. **6** is a graph representing reduction percentages (%) of reboiler heat duty in the absorbent regenerator in test example 1. FIG. **6** represents the reduction percentages (%) calculated based on standardized reboiler heat duty reduction percentages (0) without the discharge.

As illustrated in FIGS. **4** to **6**, in test example 1, the separation flow ratio in the range of 1:1 to 60:1 allows for improvement in the separation effects on the CO_2 concentration ratio in the rich solution, the CO_2 recovery amount ratio, and the reboiler heat duty reduction percentage (%). In particular, the separation flow ratio in the range of 1:1 to 40:1 allows for improvement in the increase ratio of the separation effects on the CO_2 concentration ratio in the rich solution, the CO_2 recovery amount ratio, and the reboiler heat duty reduction percentage (%).

Here, the CO_2 absorption rate of the absorbent is represented as the product of the volumetric mass transfer coefficient (the product of the mass transfer coefficient and the gas-liquid contact area) and the driving force (the difference between the CO_2 partial pressure and the CO_2 equilibrium pressure). In the case of returning the absorbent of the semi-rich solution **13B** (circulation solution **13B**) circulating from the rich solution side at the bottom side of the CO_2 absorber **14** to the top of the CO_2 absorber **14**, as disclosed in the prior art (Patent Literature 1), the concentration of acid gas (CO_2) may be increased at the outlet side due to decrease in the driving force when the CO_2 content is high in the gas (such as boiler flue gas with a CO_2 concentration of more than 10% and 15% or less).

In contrast, when CO_2 is removed from the gas with a low CO_2 concentration of 1% such as atmospheric air as in this test example, even if the absorbent is circulated from the rich solution side on the bottom side of the CO_2 absorber to the lean solution side on the upper side of the CO_2 absorber including the top thereof, it is possible to improve the volumetric mass transfer coefficient to such a great extent as

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to outweigh the decrease of the driving force, and thereby improve the CO₂ absorption rate. Thus, the improvement in CO₂ absorption rate in the present invention is particularly noticeable under conditions with the relatively low CO₂ concentration in the target gas.

Therefore, examples of the gas with low CO₂ concentration applicable to the present invention include a gas at nearly atmospheric pressure containing a subtle amount (with a CO₂ concentration of 1% or less), preferably a gas at nearly atmospheric pressure with a CO₂ concentration of 10% or less, more preferably a gas at nearly atmospheric pressure with a CO₂ concentration of 5% or less. Examples of the gas with a CO₂ concentration of 10% or less include flue gases from boilers, gas turbines, combustion furnaces, heating furnaces, incinerators, internal combustion engines, and the like, for example, as well as atmospheric air and air in closed and substantially closed spaces.

In the present invention, in the case of targeting air, or air in the closed or substantially closed space, the lean absorbent temperature is desired to be same as or as close as possible to the target gas temperature.

When the introduced gas 11 in the CO₂ absorber 14 is atmospheric gas or air with unsaturated water vapor, the water content of the absorbent is evaporated and then exhausted as moisture saturated gas, thereby requiring water to be supplied to the absorbent. The higher the lean solution temperature, the higher amount of water supplied to the absorbent, due to increase in the temperature of the outlet gas 11C of the CO₂ absorber 14 and increased amount of exhausted water content resulting from increase of saturated water content. For this reason, as illustrated in FIG. 1, the lean solution supply line L₂ is provided with a supply water introduction line L₉ to introduce supplied water 70 to dilute the absorbent in the present embodiment, preventing the amine concentration from increased in the absorbent.

The amount of absorbed CO₂ is small when the introduced gas 11A in the CO₂ absorber 14 is atmospheric gas or air with unsaturated water vapor. Thus, the amount of heat absorption resulting from the evaporative latent heat of water content is greater than the amount of heat generated by the reaction when CO₂ is absorbed from the flue gas exhausted from the boiler as in the conventional system. Therefore, even when the lean solution 13A with the same temperature as the introduced gas 11A is used, for example, the outlet gas temperature (T₂) of the outlet gas 11C is lower by approximately 3° C. than that (T₁) of the introduced gas 11A under the condition that the gas temperature (T₁) of the introduced gas 11A in test example 1 according to first embodiment is 25° C.

Thus, under the condition that CO₂ is removed from a gas containing a subtle amount of CO₂ (a CO₂ concentration of 1% or less), the temperature of the circulation solution in the absorption section is lower than that of the lean solution 13A introduced into the CO₂ absorber 14, without a cooler installed in the first absorbent circulation line L₁₁ and the second absorbent circulation line L₁₂. As a result, the reaction heat of the amine solution is not generated when CO₂ is absorbed inside the absorber as in the prior arts, thereby not requiring the cooler in the first absorbent circulation line L₁₁ and the second absorbent circulation line L₁₂ in which the semi-rich solution 13B and the rich solution 13C are circulated.

FIGS. 7 to 9 illustrate performances with comparison to the prior art (Patent Literature 2: International Publication No. 09/104744) in which only the second absorbent circulation line L₁₂ is used to circulate the absorbent, without the first absorbent circulation line L₁₁ installed as in the present

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invention, (with circulation at the lower stage without circulation at the upper stage) (Comparative Example 2).

FIG. 7 is a graph representing CO₂ concentration ratios in the rich solution in test example 2. Herein, FIG. 7 represents a performance improvement ratio calculated based on standardized CO₂ concentration (1) with circulation at the lower stage without circulation at the upper stage. In FIG. 7, the horizontal axis represents the separation flow ratio of the flow rate (f1) of the semi-rich solution 13B (circulation solution 13B) circulated through the first absorbent circulation line L₁₁ to the flow rate (f2) of the discharged solution 13B-1 supplied to the absorbent discharge line L₂₁, in the range of 1:1 to 60:1 (the same applies to FIGS. 9 and 10). The same applies to the separation flow ratio of the flow rate (f1) of the circulation solution 13C-1 circulated through the second absorbent circulation line L₁₂ to the flow rate (f2) of the rich solution 13C supplied to the rich solution supply line L₁, in the range of 1:1 to 60:1.

FIG. 8 is a graph representing CO₂ recovery amount ratios in test example 2. Herein, FIG. 8 represents the performance improvement ratio calculated based on standardized CO₂ recovery amount (1) with circulation at the lower stage without circulation at the upper stage.

FIG. 9 is a graph representing reduction percentages (%) of reboiler heat duty in the absorbent regenerator in test example 2. FIG. 9 represents the reduction percentages (%) calculated based on standardized reboiler heat duty reduction percentages (0) with circulation at the lower stage without circulation at the upper stage.

As illustrated in FIGS. 7 to 9, in test example 2, the separation ratio of the flow rates in the range of 1:1 to 60:1 allows for improvement in the separation effects on the CO₂ concentration ratio in the rich solution, the CO₂ recovery amount ratio, and the reboiler heat duty reduction percentage (%). In particular, the separation ratio of the flow rates in the range of 1:1 to 30:1 allows for improvement in the increase ratio of the separation effects on the CO₂ concentration ratio in the rich solution, the CO₂ recovery amount ratio, and the reboiler heat duty reduction percentage (%).

Although explanations are given for the CO₂ recovery unit 10A according to the present embodiment in which two absorption stages (the first absorption stage 141A and the second absorption stage 141B) are provided in the CO₂ absorber 14, the present invention is not limited to this. Two or more absorption stages are provided (another absorption stage is provided between the first absorption stage 141A and the second absorption stage 141B) in the CO₂ absorber 14 such that the circulation and discharge are performed at the predetermined separation ratio of the flow rates to circulate the absorbent.

Second Embodiment

FIG. 2 is a schematic view representing configuration of a CO₂ recovery unit according to a second embodiment. Descriptions are given for new configuration in the second embodiment added to the CO₂ recovery unit 10A according to the first embodiment illustrated in FIG. 1. The same components as in the configuration according to the first embodiment are not described.

A CO₂ recovery unit 10B according to the second embodiment is provided with a wash section 142 that is added inside the top part 14a of the first absorption stage 141A of the CO₂ absorber 14 on a downstream side of the gas flow in the CO₂ recovery unit 10A according to the first embodiment.

The wash section 142 is provided at its bottom with a liquid reservoir 144A that stores therein washing water W₂

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to wash the decarbonated outlet gas 11C. A circulation line L_3 is provided between the liquid reservoir 144A and the top of the wash section 142 to supply wash water W_2 containing the CO_2 absorbent collected in the liquid reservoir 144A for circulation from the top part 14a side of the wash section 142. The circulation line L_3 is equipped with a heat exchanger 21 that cools the wash water W_2 and a circulation pump 22 that circulates the wash water W_2 containing the CO_2 absorbent collected in the liquid reservoir 144A through the heat exchanger 21 in the circulation line L_3 .

In the present embodiment, CO_2 is recovered in the gas with low CO_2 concentration, thereby the CO_2 absorption does not involve the exothermic reaction of the amine-based absorbent, as described above. However, the water wash section 142 illustrated in FIG. 2 is preferably provided to minimize loss of the amine-based absorbent in the outlet gas 11C discharged outwardly, from the perspective of achieving near zero emission.

Third Embodiment

FIG. 3 is a schematic view representing configuration of a CO_2 recovery unit according to a third embodiment. Descriptions are given for new configuration in the third embodiment added to the CO_2 recovery unit 10B according to the second embodiment illustrated in FIG. 2. The same components as in the configuration according to the first embodiment are not described.

A CO_2 recovery unit 10C according to the third embodiment is provided with a gas quencher 12, which is added on an upstream side of the CO_2 absorber 14 in the CO_2 recovery unit 10B according to the second embodiment, to cool the introduced gas 11A down to a predetermined temperature. The gas quencher 12 has a quencher section 121 to cool the introduced gas 11A. A water circulation line L_7 is provided to circulate the cooling water W_1 between the bottom 12b side of the gas quencher 12 and the top 12a side of the quencher section 121. The water circulation line L_7 is equipped with a heat exchanger 122 that cools the cooling water W_1 and a circulation pump 123 that circulates the cooling water W_1 in the water circulation line L_7 .

In the quencher section 121, the introduced gas 11A is brought into countercurrent contact with the cooling water W_1 so as to be cooled down to a predetermined temperature. The heat exchanger 122 cools the cooling water W_1 heated by heat exchange with the introduced gas 11A. The circulation pump 123 supplies the cooling water W_1 flowing down to the bottom 12b of the gas quencher 12 via the heat exchanger 122 to the top 12a of the quencher section 121. After cooled, the introduced gas 11A is introduced into the CO_2 absorber 14 from the vicinity of the bottom of the CO_2 absorber 14 via the introduction line L_8 .

In the third embodiment, the cooling part 55, which is installed in the lean solution supply line L_2 in the preceding configuration, is installed in the first absorbent circulation line L_{11} . This allows for cooling of the mixed solution (13A/13B) of the semi-rich solution 13B (circulation solution 13B) circulating from the first absorbent circulation line L_{11} and the lean solution 13A flowed from the lean solution supply line L_2 .

The present embodiment differs from the first embodiment in that it is preferable to aim for gases (e.g., turbine flue gas and various flue gases) containing 3% or more and 10% or less of CO_2 as the introduced gas 11A. For the gases containing 3% or more of CO_2 , the liquid temperature of the lean solution 13A is expected to be a lean absorbent tem-

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perature at a temperature level of the water vapor-saturated gas flowed at the outlet of the gas quencher 12.

For example, under the condition that the gas temperature of the introduced gas 11A in the present embodiment is 40° C., when the absorbent is not circulated at each stage as in the prior art, the gas temperature of the outlet gas 11C discharged from the top 14a is approximately 10° C. higher than the gas temperature of the introduced gas. Therefore, under such a condition, it is more advantageous to install the cooling part 55 in the first absorbent circulation line L_{11} to prevent the temperature of the absorbent from rising, for increasing the CO_2 concentration in the absorbent.

Hereinafter, preferred test examples exhibiting effects of the separation flow ratio according to the present embodiment are described. The present invention is not limited to these examples. FIGS. 10 to 15 of the test examples illustrate examples aimed for gases with a CO_2 concentration of 3% or more.

FIG. 10 is a graph representing CO_2 concentration ratios in the rich solution in test example 3. Herein, FIG. 10 represents a performance improvement ratio calculated based on standardized CO_2 concentration (1) without the discharge at each stage. In FIG. 10, the horizontal axis represents the separation flow ratio of the flow rate (f1) of the semi-rich solution 13B (circulation solution 13B) circulated through the first absorbent circulation line L_{11} to the flow rate (f2) of the discharged solution 13B-1 supplied to the absorbent discharge line L_{21} , in the range of 1:1 to 60:1 (the same applies to FIGS. 11 and 12). The same applies to the separation flow ratio of the flow rate (f1) of the circulation solution 13C-1 circulated through the second absorbent circulation line L_{12} to the flow rate (f2) of the rich solution 13C supplied to the rich solution supply line L_1 , in the range of 1:1 to 60:1.

FIG. 11 is a graph representing CO_2 recovery amount ratios in test example 3. Herein, FIG. 11 represents a performance improvement ratio calculated based on standardized CO_2 recovery amount (1) without the discharge at each stage. FIG. 12 is a graph representing reduction percentages (%) of reboiler heat duty in the absorbent regenerator in test example 3. FIG. 13 represents the reduction percentages (%) calculated based on standardized reboiler heat duty reduction percentages (0) without the discharge.

As illustrated in FIGS. 10 to 12, in test example 3, the separation flow ratio in the range of 1:1 to 60:1 allows for increase of the separation effects on the CO_2 concentration ratio in the rich solution, the CO_2 recovery amount ratio, and the reboiler heat duty reduction percentage (%). In particular, the separation flow ratio in the range of 1:1 to 20:1 allows for improvement in the increase ratio of the separation effects on the CO_2 concentration ratio in the rich solution, the CO_2 recovery amount ratio, and the reboiler heat duty reduction percentage (%).

FIGS. 13 to 15 illustrate performances with comparison to the prior art (Patent Literature 2: International Publication No. 09/104744) in which only the second absorbent circulation line L_{12} is used to circulate the absorbent, without the first absorbent circulation line L_{11} installed as in the present invention, (with circulation at the lower stage without circulation at the upper stage) (Comparative Example 2).

FIG. 13 is a graph representing CO_2 concentration ratios in the rich solution in test example 4. Herein, FIG. 13 represents a performance improvement ratio calculated based on standardized CO_2 concentration (1) with circulation at the lower stage without circulation at the upper stage. In FIG. 7, the horizontal axis represents the separation flow ratio of the flow rate (f1) of the semi-rich solution 13B

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(circulation solution **13B**) circulated through the first absorbent circulation line L_{11} to the flow rate (f2) of the discharged solution **13B-1** supplied to the absorbent discharge line L_{21} , in the range of 1:1 to 60:1 (the same applies to FIGS. **14** and **15**). The same applies to the separation flow ratio of the flow rate (f1) of the circulation solution **13C-1** circulated through the second absorbent circulation line L_{12} to the flow rate (f2) of the rich solution **13C** supplied to the rich solution supply line L_1 , in the range of 1:1 to 60:1.

FIG. **14** is a graph representing CO_2 recovery amount ratios in test example 4. Herein, FIG. **14** represents the performance improvement ratio calculated based on standardized CO_2 recovery amount (1) with circulation at the lower stage without circulation at the upper stage. FIG. **15** is a graph representing reduction percentages (%) of reboiler heat duty in the absorbent regenerator in test example 4. FIG. **15** represents the reduction percentages (%) calculated based on standardized the reboiler heat duty reduction percentages (0) with circulation at the lower stage without circulation at the upper stage.

As illustrated in FIGS. **13** to **15**, in test example 4, the separation ratio of the flow rates in the range of 1:1 to 60:1 allows for improvement of the separation effects on the CO_2 concentration ratio in the rich solution, the CO_2 recovery amount ratio, and the reboiler heat duty reduction percentage (%). In particular, the separation ratio of the flow rates in the range of 1:1 to 40:1 allows for improvement in the increase ratio of the separation effects on the CO_2 concentration ratio in the rich solution, the CO_2 recovery amount ratio, and the reboiler heat duty reduction percentage (%).

The CO_2 recovery unit and the CO_2 recovery method described in the embodiments are understood, for example, as follows.

The CO_2 recovery unit **10A**, **10B**, **10C** according to a first aspect includes: the CO_2 absorber **14** configured to bring the introduced gas **11A** having a low CO_2 concentration into countercurrent contact with the CO_2 absorbent and remove CO_2 ; the absorbent regenerator **15** configured for heat exchange between the rich solution **13C** containing absorbed CO_2 and the vapor flowed from the reboiler **31** to regenerate the CO_2 absorbent; the rich solution supply line L_1 configured to discharge the rich solution **13C** containing absorbed CO_2 from the bottom **14b** of the CO_2 absorber **14** and supply the rich solution to the top side of the absorbent regenerator **15**; and the lean solution supply line L_2 configured to discharge the lean solution **13A**, which is obtained after releasing CO_2 , from the bottom **15b** of the absorbent regenerator **15**, and supply it to the top side of the CO_2 absorber **14**, in which the CO_2 absorber **14** has at least two or more CO_2 absorption stages **141A**, **141B**. The CO_2 recovery unit **10A** further includes: the absorbent circulation lines L_{11} , L_{12} configured to discharge the CO_2 absorbent from the lower sides of the CO_2 absorption stages **141A**, **141B** and supply the CO_2 absorbent as the circulating semi-rich solution **13B** (circulation solution **13B**) and the circulating rich solution **13C** (circulation solution **13C-1**) to the upper sides of the corresponding CO_2 absorption stages **141A**, **141B** that are used as the discharge stages; and the absorbent discharge lines L_{21} , L_1 configured to discharge parts of the semi-rich solution **13B** (circulation solution **13B**) circulation solution **13C-1** from the absorbent circulation lines L_{11} , L_{12} and supply the solution as the discharged solutions **13B-1**, **13C** to adjacent stage immediately below the discharge stages.

This configuration enables it to discharge the CO_2 absorbent containing the absorbed CO_2 from the lower portion of each of the CO_2 absorption stages, supply it to the upper portion of the corresponding stage at which the CO_2 absor-

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bent is discharged, and reuse the solution as the CO_2 absorbent for the purpose of recovering CO_2 in the gas, increasing the CO_2 concentration in the CO_2 absorbent, and reducing the reboiler heat duty needed for recovering CO_2 from a variety of gases with low CO_2 contents.

The CO_2 recovery unit **10A**, **10B**, **10C** according to a second aspect is aimed for the introduced gas **11A** with a CO_2 concentration of 10% by volume or less.

This configuration enables it to reduce the reboiler heat duty needed for recovering CO_2 from the introduced gas **11A** with a CO_2 concentration of 10% by volume or less.

In the CO_2 recovery unit **10A**, **10B**, **10C** according to a third aspect, the flow ratio is set at 1:1 to 60:1 for the separation ratio of the flow rates of the circulation solution **13B**, **13C-1** to the discharged solution **13B-1**, **13C**.

With the separation ratio of the flow rates of the circulation solution **13B**, **13C-1** to the discharged solution **13B-1**, **13C** ranging from 1:1 to 60:1 in this configuration, it is possible to reduce the reboiler heat duty needed for recovering CO_2 from a variety of gases with low CO_2 contents.

The CO_2 recovery unit **10B**, **10C** according to a fourth aspect is provided with the wash section **142** at the downstream side of gas flow in the CO_2 absorption section at the highest stage.

This configuration having the wash section **142** enables it to wash the outlet gas **11C** with water, and thereby minimize the loss of the amine-based absorbent in the outlet gas **11C** released outwardly.

The CO_2 recovery unit **10C** according to a fifth aspect has the cooling part **55** in the first absorbent circulation line L_{11} .

With the cooling part **55** in the first absorbent circulation line L_{11} , this configuration enables it to prevent the temperature of the absorbent from increasing, achieving advantageous effects on the increase in the CO_2 concentration in the absorbent.

A CO_2 recovery method according to a sixth aspect includes: a CO_2 absorption step of bringing the introduced gas **11A** having low CO_2 concentration into countercurrent contact with the CO_2 absorbent to remove CO_2 ; an absorbent regeneration step of performing the heat exchange between the rich solution **13C** containing the absorbed CO_2 and the steam flowed from the reboiler **31** to regenerate the CO_2 absorbent; and a step of discharging the rich solution **13C** containing CO_2 that is absorbed at the CO_2 absorption step, supplying it to the absorbent regeneration step, discharging the lean solution **13A** that is obtained by releasing CO_2 at the absorbent regeneration step, and supplying it to the CO_2 absorption step to recover CO_2 in the gas. The CO_2 absorption step includes an absorbent circulation step of, in the CO_2 absorption section with at least two or more stages **141A**, **141B**, discharging the CO_2 absorbent from the lower sides of the CO_2 absorption stages **141A**, **141B** and then supplying it as the circulation solution **13B**, **13C-1** to the upper sides of the corresponding CO_2 absorption stages **141A**, **141B** that are used for the discharge; and an absorbent discharge step of discharging a part of the circulation solution **13B** from the absorbent circulation step and supplying it as the discharged solution **13B-1**, **13C** to the stage immediately below the stage used for the discharge.

This configuration enables it to discharge the CO_2 absorbent containing the absorbed CO_2 from the lower portion of each of the CO_2 absorption stages, supply it to the upper portion of the corresponding stage at which the CO_2 absorbent is discharged, and reuse the solution as the CO_2 absorbent for the purpose of recovering CO_2 in the gas, increasing the CO_2 concentration in the CO_2 absorbent, and

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reducing the reboiler heat duty needed for recovering CO₂ from a variety of gases with low CO₂ contents.

REFERENCE SIGNS LIST

10A to 10C CO₂ recovery unit
 11A, 11B Introduced gas
 11C Outlet gas
 13A Lean solution
 13B Semi-rich solution
 13B-1 Discharged solution
 13C Rich solution
 13C-1 Circulation solution
 14 CO₂ absorber
 15 Absorbent regenerator
 31 Reboiler
 55 Cooling part
 141A First CO₂ absorption stage (first absorption stage)
 141B Second CO₂ absorption stage (second absorption stage)
 144 Liquid reservoir
 L₁ Rich solution supply line
 L₂ Lean solution supply line
 L₁₁ First absorbent circulation line
 L₁₂ Second absorbent circulation line
 L₂₁ Absorbent discharge line

The invention claimed is:

1. A CO₂ recovery unit comprising:
 - a CO₂ absorber that brings a gas having a low CO₂ concentration into countercurrent contact with a CO₂ absorbent to remove CO₂ from the gas;
 - an absorbent regenerator that performs a heat exchange between a rich solution containing absorbed CO₂ and water vapor flowed from a reboiler to regenerate the CO₂ absorbent;
 - a rich solution supply line that discharges a rich solution in which CO₂ has been absorbed by the CO₂ absorbent in the CO₂ absorber, from a bottom part of the CO₂ absorber and supplies the rich solution to an upper side of the absorbent regenerator; and
 - a lean solution supply line that discharges a lean solution in which CO₂ has been released from the rich solution in the absorbent regenerator, from a bottom part of the absorbent regenerator and supplies the lean solution as the CO₂ absorbent to an upper side of the CO₂ absorber, wherein the CO₂ absorber includes at least two or more CO₂ absorption sections including a first CO₂ absorption section and a second CO₂ absorption section located below the first CO₂ absorption section, and the CO₂ recovery unit further comprises:
 - a first absorbent circulation line that discharges the CO₂ absorbent from the first CO₂ absorption section and supplies the CO₂ absorbent as a first circulation solution to an upper side of the first CO₂ absorption section;

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- a second absorbent circulation line that discharges the CO₂ absorbent from the second CO₂ absorption section and supplies the CO₂ absorbent as a second circulation solution to an upper side of the second CO₂ absorption section; and
 - an absorbent discharge line that discharges a part of the first circulation solution from the first absorbent circulation line and supply the part of the first circulation solution as a discharged solution to the second absorbent circulation line.
2. The CO₂ recovery unit according to claim 1, wherein the gas has a CO₂ concentration of 10% by volume or less.
 3. The CO₂ recovery unit according to claim 1, wherein a separation ratio of flow rates of the first circulation solution to the discharged solution is set to 1:1 to 60:1.
 4. The CO₂ recovery unit according to claim 1, further comprising a water wash section on a downstream side of a gas flow in the first CO₂ absorption section.
 5. The CO₂ recovery unit according to claim 1, wherein the first absorbent circulation line includes a cooling part.
 6. A CO₂ recovery method comprising:
 - a CO₂ absorption step of bringing a gas having a low CO₂ concentration into countercurrent contact with a CO₂ absorbent to remove CO₂ from the gas;
 - an absorbent regeneration step of performing a heat exchange between water vapor and a rich solution containing absorbed CO₂ to regenerate the CO₂ absorbent; and
 - a step of discharging a rich solution in which CO₂ has been absorbed at the CO₂ absorption step, supplying the rich solution to the absorbent regeneration step, discharging a lean solution in which CO₂ has been released from the rich solution at the absorbent regeneration step, and supplying the lean solution as the CO₂ absorbent to the CO₂ absorption step to recover CO₂ in the gas,
 wherein the CO₂ absorption step includes at least two or more CO₂ absorption steps including a first CO₂ absorption step and a second CO₂ absorption step following the first CO₂ absorption step,
 - the CO₂ recovery method further comprises:
 - a first absorbent circulation step of discharging the CO₂ absorbent obtained at the first CO₂ absorption step and supplying the CO₂ absorbent as a first circulation solution to the first CO₂ absorption step;
 - a second absorbent circulation step of discharging the CO₂ absorbent obtained at the second CO₂ absorption step and supplying the CO₂ absorbent as a second circulation solution to the second CO₂ absorption step; and
 - an absorbent discharge step of discharging a part of the first circulation solution obtained at the first absorbent circulation step and supplying the part of the first circulation solution as a discharged solution to the second absorbent circulation step.

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